CBSE BOARD EXAMINATION 2022

MATHEMATICS (STANDARD) — TERM II

Class 10th — Solved Paper with Answers

Maximum Marks:	40 (Term II)
Time Allowed:	2 Hours
Date of Exam:	May 5, 2022
Total Questions:	14 (All Compulsory)
Sections:	A (MCQs 1M), B (SA 2M), C (SA 3M), D (LA 5M)

GENERAL INSTRUCTIONS:

(i) This question paper contains 14 questions. All questions are compulsory.

(ii) This paper is divided into four sections — A, B, C and D.

(iii) Section A: Q1–Q6 are Very Short Answer type questions (2 marks each).

(iv) Section B: Q7–Q10 are Short Answer type questions (3 marks each).

(v) Section C: Q11–Q12 are Long Answer type questions (5 marks each).

(vi) Section D: Q13–Q14 are Case Study based questions (4 marks each).

(vii) Internal choice is provided in some questions.

(viii) Draw neat diagrams wherever required. Take $\pi = 22/7$ wherever required.

(ix) Use of calculators is NOT allowed.

SECTION A — VERY SHORT ANSWER QUESTIONS (2 Marks Each)

Q1. Find the value of k for which the system of equations x + 2y = 5 and 3x + ky + 15 = 0 has no solution.

Answer:
For no solution: a■/a■ = b■/b■ ≠ c■/c■
1/3 = 2/k → k = 6
Check: c■/c■ = 5/15 = 1/3
Since a■/a■ = b■/b■ = c■/c■ when k=6, this gives infinite solutions.
For NO solution: we need a■/a■ = b■/b■ ≠ c■/c■
So k = 6 is rejected. Actually for no solution with these equations: 1/3 = 2/k ≠ 5/15
This is not possible. [Note: Question may vary by set]
For standard form: If equations are x + 2y – 5 = 0 and 3x + ky + 15 = 0
Then c■/c■ = –5/15 = –1/3 ≠ 1/3
So k = **6** gives no solution (since 1/3 = 2/6 ≠ –5/15)

Solution: No solution ke liye a■/a■ = b■/b■ ≠ c■/c■ hona chahiye.

Q2. Find the sum of first 20 terms of the A.P.: –5, –2, 1, 4, ...

Answer:
a = –5, d = 3
S■■ = (20/2)[2(–5) + 19(3)] = 10[–10 + 57] = 10 × 47 = **470**

Solution: Sum formula: S■ = n/2[2a + (n–1)d].

Q3. (OR) If α and β are the zeroes of the polynomial $x^2 - 5x + 6$, find the value of $\alpha^2 + \beta^2$.

Answer:
$\alpha + \beta = 5$, $\alpha\beta = 6$
$\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta = 25 - 12 =$ **13**

Solution: $\alpha^2 + \beta^2 = (\alpha+\beta)^2 - 2\alpha\beta$ identity use karo.

Q4. Find the coordinates of the point which divides the line segment joining (–2, 3) and (4, –5) in the ratio 2:3 internally.

Answer:
x = (2×4 + 3×(–2))/(2+3) = (8–6)/5 = 2/5
y = (2×(–5) + 3×3)/5 = (–10+9)/5 = –1/5
Point = (2/5, –1/5)

Solution: Section formula: x = (m■x■ + m■x■)/(m■+m■), y = (m■y■ + m■y■)/(m■+m■).

Q5. Prove that √3 is an irrational number.

Answer:
Proof: Assume √3 is rational.
√3 = a/b where a, b are coprime.
$3 = a^2/b^2 \rightarrow a^2 = 3b^2 \rightarrow$ 3 divides $a^2 \rightarrow$ 3 divides a.
Let a = 3c. Then $9c^2 = 3b^2 \rightarrow b^2 = 3c^2 \rightarrow$ 3 divides b.
Contradiction! Hence **√3 is irrational. (Proved)**

Solution: Contradiction method. Assume rational, show 3 divides both a and b.

Q6. In the figure, DE || BC. If AD = 2 cm, DB = 3 cm and AE = 1.6 cm, find EC.

Answer:
By BPT: AD/DB = AE/EC
2/3 = 1.6/EC → EC = (1.6 × 3)/2 = **2.4 cm**

Solution: Basic Proportionality Theorem: AD/DB = AE/EC.

SECTION B — SHORT ANSWER QUESTIONS (3 Marks Each)

Q7. Solve the following pair of linear equations: $2x + 3y = 11$ and $2x - 4y = -24$. Hence find the value of 'm' for which $y = mx + 3$.

Answer:
$2x + 3y = 11$ ... (i)
$2x - 4y = -24$ ... (ii)
Subtract (ii) from (i): $7y = 35 \rightarrow y = 5$
From (i): $2x + 15 = 11 \rightarrow 2x = -4 \rightarrow x = -2$
Solution: $x = -2, y = 5$

Put in $y = mx + 3$: $5 = m(-2) + 3 \rightarrow -2m = 2 \rightarrow$ $\mathbf{m = -1}$

Solution: Elimination method. Equations subtract kar ke y nikaalo, fir x, fir m.

Q8. (OR) Find the HCF and LCM of 180 and 252 using prime factorisation method.

Answer:
$180 = 2^2 \times 3^2 \times 5$
$252 = 2^2 \times 3^2 \times 7$
HCF = $2^2 \times 3^2$ = **36**
LCM = $2^2 \times 3^2 \times 5 \times 7$ = **1260**

Solution: Prime factorisation se common factors HCF ke liye, saare factors LCM ke liye.

Q9. Prove that: $(\sin\theta + \csc\theta)^2 + (\cos\theta + \sec\theta)^2 = 7 + \tan^2\theta + \cot^2\theta$.

Answer:
LHS: $(\sin\theta + 1/\sin\theta)^2 + (\cos\theta + 1/\cos\theta)^2$
$= \sin^2\theta + 2 + 1/\sin^2\theta + \cos^2\theta + 2 + 1/\cos^2\theta$
$= (\sin^2\theta + \cos^2\theta) + 4 + \csc^2\theta + \sec^2\theta$
$= 1 + 4 + (1 + \cot^2\theta) + (1 + \tan^2\theta)$
$= 7 + \tan^2\theta + \cot^2\theta$ = **RHS (Proved)**

Solution: Expand karo. $\sin^2\theta + \cos^2\theta = 1$. $\csc^2\theta = 1 + \cot^2\theta$, $\sec^2\theta = 1 + \tan^2\theta$ use karo.

Q10. A chord of a circle of radius 10 cm subtends a right angle at the centre. Find the area of the corresponding minor segment. (Use $\pi = 3.14$)

Answer:
Given: $r = 10$ cm, $\theta = 90°$
Area of sector = $(90/360) \times 3.14 \times 100 = 78.5$ cm²
Area of $\Delta = \frac{1}{2} \times 10 \times 10 = 50$ cm²
Area of minor segment = $78.5 - 50$ = **28.5 cm²**

Solution: Segment = Sector – Triangle. 90° wala triangle right-angled isosceles hota hai.

SECTION C — LONG ANSWER QUESTIONS (5 Marks Each)

Q11. (a) Prove that if a line is drawn parallel to one side of a triangle to intersect the other two sides in distinct points, then the other two sides are divided in the same ratio.
(OR) (b) In the figure, PA, QB and RC are each perpendicular to AC. If AP = x, BQ = y and CR = z, then prove that 1/x + 1/z = 1/y.

Answer:
(a) Proof of Basic Proportionality Theorem (BPT/Thales Theorem):
Given: ΔABC with DE || BC, intersecting AB at D and AC at E.
To prove: AD/DB = AE/EC
Construction: Join BE and CD. Draw EF ⊥ AB and DG ⊥ AC.
Area(ΔADE) = ½ × AD × EF
Area(ΔBDE) = ½ × DB × EF
∴ Area(ΔADE)/Area(ΔBDE) = AD/DB ... (i)
Similarly, Area(ΔADE)/Area(ΔCDE) = AE/EC ... (ii)
But Area(ΔBDE) = Area(ΔCDE) [same base DE, between same parallels]
From (i) and (ii): **AD/DB = AE/EC (Proved)**

(OR) (b) Since PA || BQ || CR (all ⊥ AC)
In ΔAPC, BQ || AP: ΔBQC ~ ΔAPC → BQ/AP = QC/AC → y/x = QC/AC
In ΔARC, BQ || CR: ΔABQ ~ ΔARC → BQ/CR = AQ/AC → y/z = AQ/AC
Adding: y/x + y/z = (QC + AQ)/AC = AC/AC = 1
y(1/x + 1/z) = 1 → **1/x + 1/z = 1/y (Proved)**

Solution: BPT ka standard proof. OR mein similar triangles use karo.

Q12. A motorboat whose speed is 18 km/h in still water takes 1 hour more to go 24 km upstream than to return downstream to the same spot. Find the speed of the stream.
(OR) The difference between the outer and inner radii of a hollow right circular cylinder of length 14 cm is 1 cm. If the volume of the metal used is 176 cm³, find the outer and inner radii.

Answer:
(a) Let speed of stream = x km/h
Speed upstream = (18 – x) km/h
Speed downstream = (18 + x) km/h
Time upstream = 24/(18–x), Time downstream = 24/(18+x)
Given: 24/(18–x) – 24/(18+x) = 1
$48x = 324 - x^2 \rightarrow x^2 + 48x - 324 = 0$
(x–6)(x+54) = 0 → x = 6
Speed of stream = 6 km/h

(OR) Let outer radius = R, inner = r
R – r = 1 ... (i)
$\pi(R^2 - r^2) \times 14 = 176 \rightarrow \pi(R-r)(R+r) \times 14 = 176$
(22/7) × 1 × (R+r) × 14 = 176 → 44(R+r) = 176 → R + r = 4 ... (ii)
From (i) and (ii): R = 2.5 cm, r = **1.5 cm**

Solution: Upstream/Downstream speed se equation banao. OR mein cylinder volume formula use karo.

SECTION D — CASE STUDY BASED QUESTIONS (4 Marks Each)

Q13. Case Study: Circus Tent

A circus tent is in the shape of a cylinder surmounted by a conical top. The height of the cylindrical part is 10 m and the height of the conical part is 5 m. The diameter of the base of the tent is 28 m.

Based on the above information, answer the following questions:
(i) Find the slant height of the conical part. (1 mark)
(ii) Find the curved surface area of the cylindrical part. (1 mark)
(iii) Find the total area of the canvas used to make the tent. (2 marks)

Answer:
(i) Radius $r = 14$ m, height of cone $h = 5$ m
Slant height $l = \sqrt{(r^2 + h^2)} = \sqrt{(196 + 25)} = \sqrt{221} \approx$ **14.87 m**

(ii) CSA of cylinder $= 2\pi rh = 2 \times (22/7) \times 14 \times 10 =$ **880 m²**

(iii) CSA of cone $= \pi rl = (22/7) \times 14 \times \sqrt{221} = 44 \times 14.87 \approx 654.28$ m²
Total canvas area $= 880 + 654.28 \approx$ **1534.28 m²**

Solution: Cylinder + Cone combination. Slant height by Pythagoras. CSA formulas use karo.

Q14. Case Study: Student Marks Distribution

The following distribution shows the marks obtained by 50 students in a Mathematics test:
Marks: 0-10, 10-20, 20-30, 30-40, 40-50
Students: 5, 8, 15, 12, 10

Based on the above information, answer the following questions:
(i) Find the modal class. (1 mark)
(ii) Find the median class. (1 mark)
(iii) Calculate the median marks. (2 marks)

Answer:
(i) Modal class = **20-30** (highest frequency = 15)

(ii) $N = 50$, $N/2 = 25$
cf: 5, 13, 28, 40, 50
25 lies in cf = 28 → Median class = **20-30**

(iii) Median $= l + [(N/2 - cf)/f] \times h$
$l = 20$, $h = 10$, $f = 15$, $cf = 13$
Median $= 20 + [(25-13)/15] \times 10 = 20 + (12/15) \times 10 = 20 + 8 =$ **28 marks**

Solution: Modal class = highest frequency. Median class jisme N/2 aata hai. Formula se median nikaalo.

— END OF SOLVED PAPER —

Best of Luck for your Exams!

This solved paper is prepared based on CBSE Class 10 Mathematics (Standard) Board Exam 2022 (Term II).
Questions and answers are compiled from official sources and expert solutions.
For any discrepancies, refer to the official CBSE answer key.